MUV PROBLEM 2

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Lab-II problem-2 **Date: 10-02-2026**

The data on weight, height and chest circumference of 15 UG students are given below.

Weight(kg)	Height(cm)	Chest circumference(cm)
60	155	36
61	158	45
54	155	39
49	146	34
63	165	46
69	170	49
67	170	46
50	150	39
57	153	45
55	151	40
59	155	41
67	171	47
63	161	41
49	150	39
50	152	40

Find the sample mean vector $\overline{X}$, population variance-covariance matrix Cov(X) and population correlation matrix Cor (X) of the above data.

TAKING VARIABLES (WEIGHT, HEIGHT AND CHEST_CIRCUMFERENCE) :

```
Weight<-c(60,61,54,49,63,69,67,50,57,55,59,67,63,49,50) #in kg
Height<-c(155,158,155,146,165,170,170,150,153,151,155,171,161,150,152) #in Cm
Chest_circumference<-c(36,45,39,34,46,49,46,39,45,40,41,47,41,39,40) #in Cm
```

- **FIRSTLY WE GOING TO FIND CORRELATION AND VAR-COVARINACE OF THE DATA MANUALLY**

VAR-COVARIANCE OF THE DATA :

```
#sample mean of the given variables(Weight,Height,chest_circumference)

mean_weight<- sum(Weight)/length(Weight)
mean_Height<- sum(Height)/length(Height)
mean_Chest_circumference<- sum(Chest_circumference)/length(Chest_circumference)

mean_weight

## [1] 58.2

mean_Height

## [1] 157.4667

mean_Chest_circumference

## [1] 41.8

mean_vector=matrix(data=c(mean_weight,mean_Height,mean_Chest_circumference),ncol=1)
mean_vector

##          [,1]
## [1,]  58.2000
## [2,] 157.4667
## [3,]  41.8000

# CO-VARIANCE OF THE POPULATION

Cov_weight_height<-(sum((Weight-mean_weight)*(Height-mean_Height)))/(15-1)
Cov_weight_chest_circum<-(sum((Weight-mean_weight)*(Chest_circumference-mean_Chest_circum
ference)))/(15-1)
Cov_height_chest_circum<-(sum((Chest_circumference - mean_Chest_circumference)*(Height-me
an_Height)))/(15-1)

Cov_height_chest_circum

## [1] 28.95714

Cov_weight_chest_circum

## [1] 23.11429

Cov_weight_height
```

```
## [1] 52.11429
```

```r
# VARIANCE OF THE POPULATION

var_weight<-(sum((Weight - mean_weight)^2))/(15-1)
var_Height<-(sum((Height - mean_Height)^2))/(15-1)
var_chest<-(sum((Chest_circumference - mean_Chest_circumference)^2))/(15-1)


var_weight

## [1] 47.31429

var_Height

## [1] 65.69524

var_chest

## [1] 18.6
```

Correlation of the Data:

```r
# co-relation of the variables

#Weight and Height

cor_weight_height<-sum((Weight - mean_weight)*(Height - mean_Height))/((sqrt((sum((Weight
-               mean_weight)^2)))*(sqrt((sum((Height - mean_Height)^2))))))


#Weight and Chest
cor_weight_chest<-sum((Weight - mean_weight)*(Chest_circumference - mean_Chest_circumfere
nce))/((sqrt((sum((Weight - mean_weight)^2)))*(sqrt((sum((Chest_circumference - mean_Ches
t_circumference)^2))))))

#Height and Chest
cor_height_chest<-sum((Height - mean_Height)*(Chest_circumference - mean_Chest_circumfere
nce))/((sqrt((sum((Height - mean_Height)^2)))*(sqrt((sum((Chest_circumference - mean_Ches
t_circumference)^2))))))


cor_height_chest

## [1] 0.8283851

cor_weight_chest

## [1] 0.7791622

cor_weight_height

## [1] 0.9347462
```

Creating a matrix :

```
final_matrix<-matrix(c(var_weight,Cov_weight_height,Cov_weight_chest_circum,
                       Cov_weight_height,var_Height,Cov_height_chest_circum,
                       Cov_weight_chest_circum,Cov_height_chest_circum,var_chest),nrow=3,
ncol=3)

colnames(final_matrix)<-c("Weight","Height","Chest_c")
rownames(final_matrix)<-c("Weight","Height","Chest_c")
final_matrix

##            Weight   Height  Chest_c
## Weight  47.31429 52.11429 23.11429
## Height  52.11429 65.69524 28.95714
## Chest_c 23.11429 28.95714 18.60000
```

Verifying the manual outcome :

```
matrices_of_variables=matrix(data=c(Weight,Height,Chest_circumference),ncol=3)

check_cov<-cov(matrices_of_variables)
colnames(check_cov)<-c("Weight","Height","Chest_c")
rownames(check_cov)<-c("Weight","Height","Chest_c")
check_cov

##            Weight   Height  Chest_c
## Weight  47.31429 52.11429 23.11429
## Height  52.11429 65.69524 28.95714
## Chest_c 23.11429 28.95714 18.60000

check_cor<-cor(matrices_of_variables)
colnames(check_cor)<-c("Weight","Height","Chest_c")
rownames(check_cor)<-c("Weight","Height","Chest_c")
check_cor

##             Weight    Height   Chest_c
## Weight  1.0000000 0.9347462 0.7791622
## Height  0.9347462 1.0000000 0.8283851
## Chest_c 0.7791622 0.8283851 1.0000000
```